

Novel quantum circuit implementation of Advanced Encryption Standard with low costs

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In this study, we examine how the quantum circuit of the Advanced Encryption Standard (AES) can be optimized from two aspects, i.e., number of qubits and T -depth. To reduce the number of qubits, we present three kinds of improved quantum circuits of S-box for different phases in the AES. We found that the number of qubits in the round function can be decreased by introducing the circuit sending $|a\rangle$ to $|S(a)\rangle$. As a result, compared with the previous quantum circuits where 400/640/768 qubits are required, our circuits of AES-128/-192/-256 only require 270/334/398 qubits. To reduce the T -depth, we propose a new circuit of AES's S-box with a T -depth of 4. Accordingly, the T -depth of our AES-128/-192/-256 quantum circuits become 80/80/84 instead of 120/120/126 in a previous study.

AES, S-box, quantum circuit, composite field arithmetic

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1 Introduction

Although the development of quantum computing is progressing, it is difficult to break the current cryptographic primitives so far. A question is when the threat of quantum computing on cryptographic primitives will become practical. Hence, it is necessary to analyze the scale of quantum computer that can break the existing cryptographic system. Moreover, the quantum resources of algorithms need to be reduced so that they can be realized on small-scale quantum computers.

Grover's search algorithm [1] can be applied to identify

the symmetric cipher key with a quadratic speed-up. The core part of this attack is Grover's oracle, which is used to evaluate whether a key is correct. Moreover, the main part of the oracle is the quantum circuit used to execute the target cipher. Thus, optimizing the quantum circuit of the target cipher is the key to reducing the quantum resource required by Grover's algorithm attack. Advanced Encryption Standard (AES) [2] is a very important symmetric cipher, and its quantum circuit implementation is drawing increasing attention [3-8].

The number of qubits in a quantum circuit is an important resource measure because the number of available qubits is limited on existing quantum computers. Many quantum circuits of AES have been designed to minimize the number of

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qubits. In 2016, Grassl et al. [3] implemented the quantum circuit of AES-128/-192/-256 with 984/1112/1336 qubits. In 2018, Almazrooie et al. [4] slightly reduced the number of qubits of AES-128 to 976. Kim et al. [5] analyzed some trade-offs between the number of qubits AND-depth in the quantum circuits of AES. In 2020, Langenberg et al. [6] gave a quantum circuit implementation of AES-128/-192/-256, which requires 864/896/1232 qubits. Then, Zou et al. [7] further reduced the number of qubits to 512/640/768. Recently, Wang et al. [9] proposed a quantum circuit of AES-128 with 400 qubits. Note that all these quantum circuits of AES are expressed with NOT, CNOT, and Toffoli gates.

The T -depth is also an important performance measure for fault-tolerant quantum circuit design as it directly determines a circuit's runtime [10]. In 2020, Jaques et al. [8] proposed a quantum circuit of AES-128/-192/-256 with a T -depth of 120/120/126. Jaques et al.'s quantum circuit of AES is expressed with Clifford+ T gates.

The realization of S-box is the key point for the quantum circuit implementation of AES because it is the only source of T gates or Toffoli gates. The classical circuits of an S-box are important for designing its quantum circuits. In a classical setting, some methods on how to efficiently calculate the S-box have been proposed [11–17]. For example, Canright [13] presented an S-box design by employing the tower field architecture over $F_{((2^2)^2)^2}$ [18], Boyar and Peralta [11,12] employed logic-minimization heuristics to reduce the gate counts and circuit depth, and Reyhani-Masoleh et al. [14] proposed an S-box design using the composite field architecture over $F_{(2^4)^2}$.

In this study, we examine the quantum circuit implementation of AES from two aspects, i.e., the number of qubits and T -depth. For simplicity, we call the circuits aiming to reduce the number of qubits (i.e., circuit width) and T -depth as W-type (“W” for width) and T-type ones, respectively. First, we present one kind of improved W-type quantum circuits of AES's S-box for $|a\rangle|0\rangle \rightarrow |a\rangle|S(a)\rangle$, $|a\rangle|b\rangle \rightarrow |a\rangle|b \oplus S(a)\rangle$ and $|a\rangle \rightarrow |S(a)\rangle$. Compared with a previous study [7], the number of qubits in our circuits is slightly reduced. Furthermore, the number of quantum gates and Toffoli depth are decreased. Second, we present a “straight-line” method in the round function of AES-128/-192/-256, which requires 142/142/142 qubits instead of 256/256/256 in a previous study [7]. Consequently, our quantum circuit of AES-128/-192/-256 requires 270/334/398 qubits instead of 400/640/768 qubits [7,9]. Third, we propose a new T-type quantum circuit implementation of AES's S-box, whose T -depth was reduced from 6 to 4 in ref. [8]. As a result, compared with a previous circuit that requires a T -depth of 120/120/126 [8], our quantum circuit of AES-128/-192/-256 only needs a T -depth of 80/80/84.

In fact, based on our AES quantum circuit, the overall attack cost of Grover's algorithm can be obtained using a straightforward manner. In particular, for the T-type one, the cost can be directly obtained by Grover's oracle code in ref. [8]. For the W-type one, it can be obtained by the Search with Two Oracles approach of ref. [19]. Considering the above straightforward manners, in this study, we only focus on the cost of realizing AES instead of the cost of running Grover's algorithm to attack AES in subsequent results.

The rest of this paper is organized as follows: In sect. 2, some preliminaries are briefly introduced. In sect. 3, some quantum circuits to prepare for the implementation of AES's S-box are given. In sects. 4 and 5, the W-type and T-type quantum circuit implementations of AES are presented, respectively. Finally, in sect. 6, a succinct conclusion is discussed.

2 Preliminaries

In this section, we briefly introduce the AES block cipher, composite field arithmetic, and quantum AND gate.

2.1 Brief description of the AES block cipher

AES [2], which is a family of iterative block ciphers based on the substitution-permutation network structure, consists of a round function and key schedule. Its three instances, namely, AES-128 (10 rounds), AES-192 (12 rounds), and AES-256 (14 rounds), correspond to key lengths of 128, 192, and 256 bits, respectively. All the three instances encrypt 128-bit plaintexts. SubBytes, ShiftRows, MixColumns, and AddRoundKey are subroutines of the round function. SubWord, RotWord, and Rcon are subroutines of the key schedule. These subroutines are simply explained as follows (see ref. [2] for more details).

SubBytes and SubWord The SubBytes transformation applies 16 S-box substitutions to 16 bytes (a byte is an 8-bit permutation) of the state, whereas the function SubWord applies four S-box substitutions to each byte of a word (a word is composed of 4 bytes). The algebraic structure of an S-box substitution is shown as follows:

The nonlinear substitution S-box takes a byte input $a \in F_{2^8} = F_2[x]/(x^8 + x^4 + x^3 + x + 1)$ and then replaces a with its multiplicative inversion a^{-1} (when $a = 0$, set $a^{-1} = 0$), followed by applying an affine transformation. Concretely, the S-box substitution can be written as:

$$S(a) = Aa^{-1} \oplus c, \quad (1)$$

where

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}.$$

ShiftRows and Rotword The ShiftRows transformation is a particular permutation of the current AES state, implemented by appropriate rewiring. Similarly, the function RotByte is a circular left shift of a word by 8 bits, also implemented by appropriate rewiring.

MixColumns The MixColumns transformation operates an entire column of the state or 4 bytes at a time, which can be specified with a 32×32 matrix. It can only be implemented by adopting CNOT gates [3,8,20] without the Toffoli gate.

AddRoundKey The AddRoundKey transformation performs the bitwise XOR of the 128-bit roundkey to the internal AES state, implemented by performing 128 CNOT gates in parallel.

Rcon The Rcon is a round constant, implemented by applying X gates (i.e., NOT gate) as appropriate.

2.2 Composite field arithmetic

2.2.1 Composite field architecture over $F_{(2^4)^2}$

Wolkerstorfer et al. [16] proposed a hardware implementation of AES's S-box using the composite fields $F_{(2^4)^2}$ and F_{2^4} :

$$\begin{cases} F_{(2^4)^2} : x^2 + x + \lambda, \\ F_{2^4} : x^4 + x + 1, \end{cases} \quad (2)$$

where $\lambda := x^3 + x^2 + x \in F_{2^4}$. Because F_{2^8} (the field mentioned in sect. 2.1) and $F_{(2^4)^2}$ are isomorphic, their mapping matrix $M : F_{2^8} \rightarrow F_{(2^4)^2}$ and the inverse matrix $M^{-1} : F_{(2^4)^2} \rightarrow F_{2^8}$ are defined as:

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}. \quad (3)$$

2.2.2 Composite field architecture over $F_{(2^2)^2}$

A tower field $F_{(2^2)^2}$ given by Boyar and Peralta [11] is shown as:

$$\begin{cases} F_{(2^2)^2} : x^2 + x + W^2, \\ F_{2^2} : x^2 + x + 1, \end{cases} \quad (4)$$

where W is the root of $x^2 + x + 1$. Based on the method mentioned in ref. [17], the mapping matrix $N : F_{2^4} \rightarrow F_{(2^2)^2}$ and inverse matrix $N^{-1} : F_{(2^2)^2} \rightarrow F_{2^4}$ are obtained as follows (here, F_{2^4} refers to the field in eq. (2)):

$$N = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad N^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}. \quad (5)$$

2.3 Quantum AND gate

The Toffoli gate can be compiled by Clifford+ T gates. A quantum AND gate has the same functionality as a Toffoli gate. However, unlike the Toffoli gate, a quantum AND gate takes $|0\rangle$ as the input state to the target qubit. Recently, Jaques et al. [8] expressed a quantum AND gate with a T -depth of 1 and implemented a quantum AND[†] gate without including any T -depth (see Figures 1 and 2 for details).

3 Proposed quantum circuit

In a classical setting, AES's S-box can be realized using a lookup table. However, to obtain an efficient quantum circuit of S-box, one can make full use of the algebraic structure

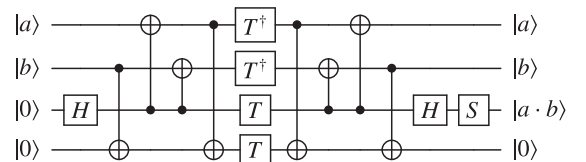


Figure 1 Quantum AND gate with T -depth 1. $T^\dagger$ is the adjoint of T .

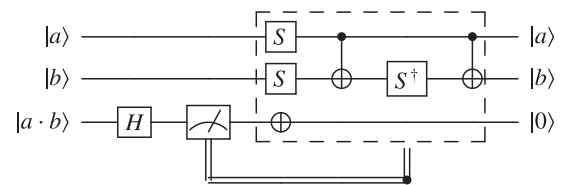


Figure 2 Quantum AND[†] gate. The dashed box indicates that if the measurement result is 1, then these operations are performed. Otherwise, the result is output directly.